



Mechanical Circulatory Support LVAD, RVAD, and Bi-VAD

History

- SAMPLE
- Bridge to transplant
- Destination therapy
- Estimated downtime
- LVAD, RVAD, Bi-Vad, TAH
- DNR, MOST, or Living Will
- Contact with LVAD coordinator

Signs and Symptoms

- Unconsciousness
- Pulseless
- Apneic
- Poor capillary refill / skin color
- AMS or decreased mental status
- No electrical activity on ECG
- No heart tones on auscultation

Differential

- See Reversible Causes below
- Infection/Sepsis
- Hypovolemia
- Cardiac arrest
- Hemorrhage

Contact VAD coordinator:

- As quickly as possible for troubleshooting and treatment advice, but do not delay emergency treatment
- Follow patient specific emergency plan if present

Rapid assessment
Check for signs of life
Assess for adequate perfusion

**Criteria for Death /
No Resuscitation**
Review DNR / MOST Form

NO

Decomposition
Rigor mortis
Dependent lividity
Blunt force trauma
Injury incompatible with life
Extended downtime with
asystole

Do not begin resuscitation

Follow
Deceased Persons
Policy

**Unresponsive and
Not breathing normally**

Assess LVAD function
Look and listen for alarms
LVAD Alarming?

NO

Place stethoscope over heart

Humming sound present?

NO

 Airway Protocol(s) AR 1, 2, 3
if indicated

 Respiratory Distress Protocol AR 4
if indicated

Assume VAD failure
Initiate age appropriate ventilation rate

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**Responsive or Unresponsive and
Breathing normally**

 Airway Protocol(s) AR 1, 2, 3
if indicated

 Respiratory Distress Protocol AR 4
if indicated

Assess LVAD function
Look and listen for alarms
LVAD Alarming?

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Go to Page 3



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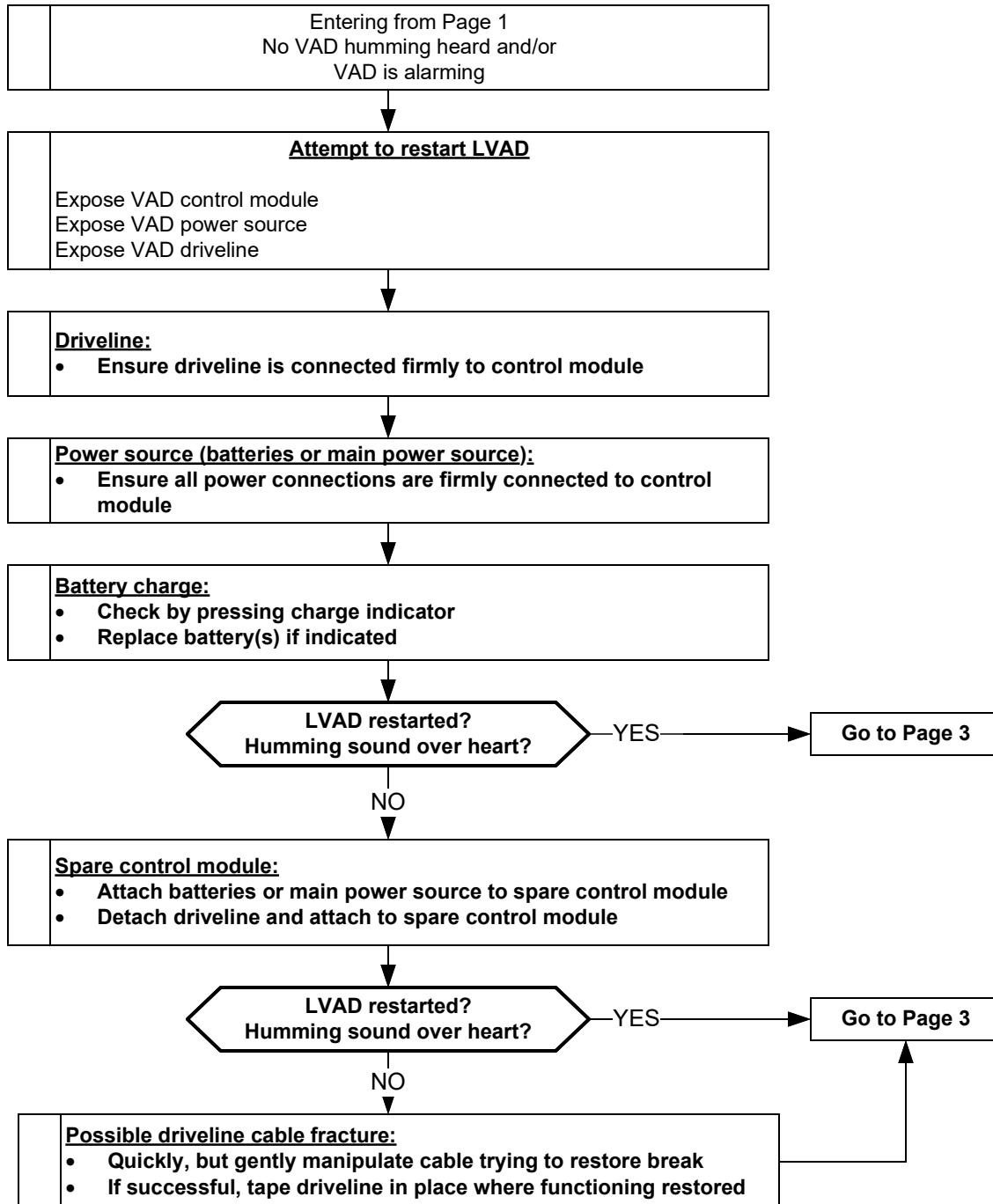
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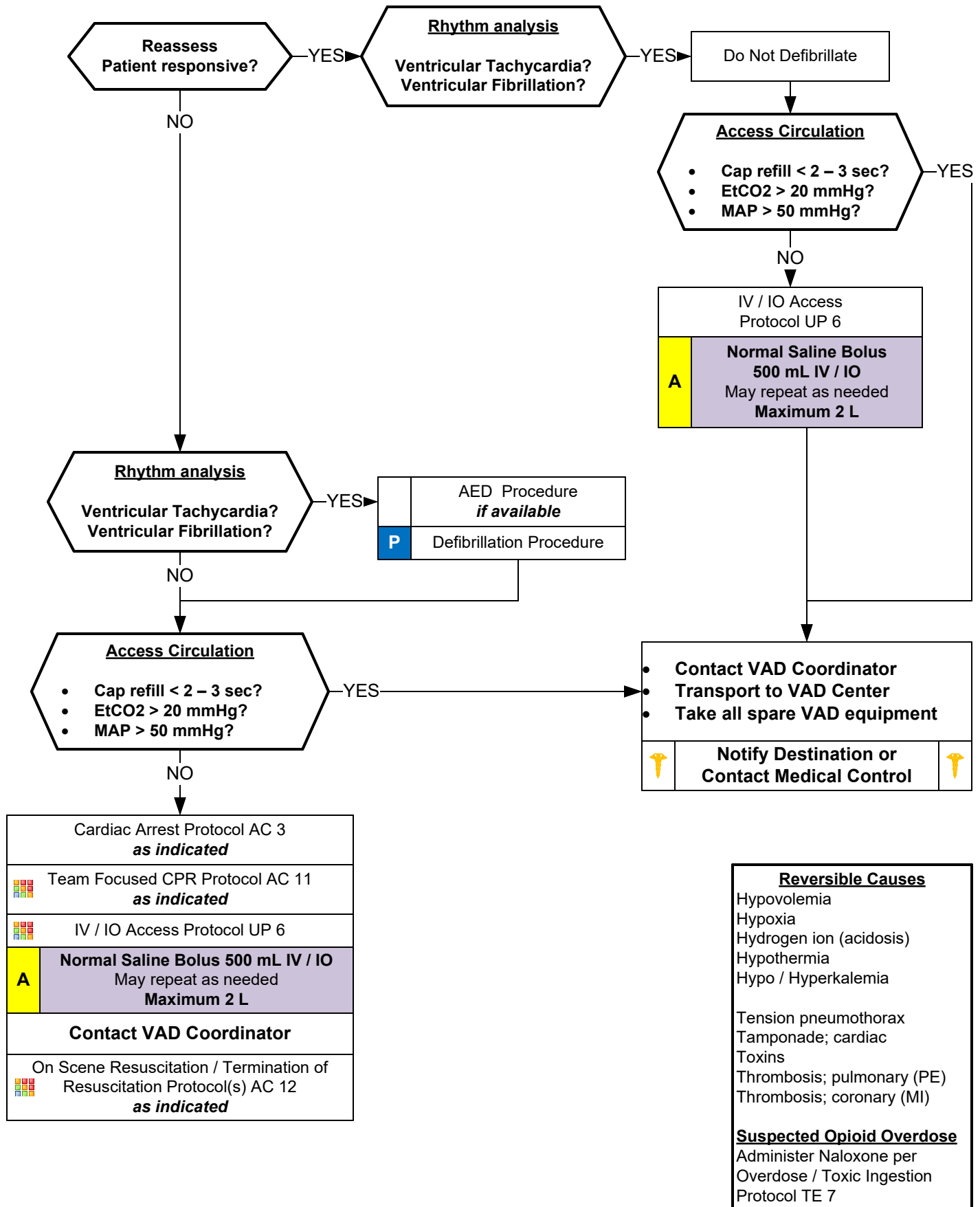
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Left Ventricular Assist Device LVAD Unresponsive or AMS

Pearls

- **Recommended exam: Mental status, skin color, capillary refill, peripheral pulses, blood pressure.**
- **Assessment of blood flow and perfusion status:**
 - Optimal BP attained by manual BP and Doppler.
 - Automated BP devices can measure a BP in about 50% of attempts and is not reliable to assess perfusion
 - A MAP of ≥ 60 mmHg is adequate for most LVAD patients.
 - Skin color, skin temperature, capillary refill
- **Mechanical Circulatory Support devices:**
 - LVAD – Left Ventricular Assist Device
 - RVAD – Right Ventricular Assist Device
 - BiVAD – Biventricular Ventricular Assist Device
 - TAH – Total Artificial Heart
- **Reasons for use:**
 - Bridge therapy – patients awaiting transplant or anticipated recovery.
 - Destination therapy – advanced heart failure, not candidate for transplant, and will live rest of life with device.
- **Pump type and assessing pulses:**
 - Pulsatile flow pumps – older units, not commonly in use now, but generate blood flow with a pulsatile flow and patient will have a palpable pulse.
 - Continuous flow pumps – majority of pumps now used and create blood flow in a continuous stream, no pulsatile flow, so patient will not have a palpable pulse.
 - Most devices are implanted inside the chest and have an internal pump, a driveline connected from the pump to the controller unit, and a power source consisting of batteries and electrical cord for receptacles.
- **Common complications:**
 - Disconnection of power supply, either battery disconnect, or electrical cord to receptacle disconnection.
 - Driveline failure or disconnection from controller unit.
 - Controller failure
 - Blood clot formation, acute stroke, and bleeding (mucosal and gastrointestinal most common sites)
 - Infection
- **Abnormal heart rhythm:**
 - Pseudo-PEA: Normal cardiac electrical activity in a patient who is alert and well perfused with no palpable pulse.
 - Tachyarrhythmias are usually well tolerated.
- **End Tidal CO₂ (EtCO₂)**
 - If EtCO₂ is < 10 mmHg, improve chest compressions. Goal is ≥ 20 mmHg.
 - If EtCO₂ spikes, typically > 40 mmHg, consider Return of Spontaneous Circulation (ROSC)
- **Transcutaneous Pacing:**
 - Pacing is NOT effective in cardiac arrest and pacing in cardiac arrest does NOT increase chance of survival